5.5.4 Real world example: the asset administration shell (AAS)

The Asset Administration Shell (AAS), developed by Plattform Industrie 4.0 and the Industrial Digital Twin Association (IDTA), is a standardized digital representation of a physical asset, crucial for Industry 4.0 digital twins.

The AAS acts as a digital "envelope" for an asset, containing all information and functionality relevant to its entire lifecycle (from design through disposal).

Its main components are:

- Submodels: Modular, standardized blocks that represent a specific aspect or function of the asset. For example:
  - A 'Digital Nameplate' for static data (serial number).
  - A 'Technical Data' submodel for specifications.
  - A 'Maintenance' submodel for service records.
- Standardized submodels ensure data is universally understandable.
- Identification and Metadata: Each AAS and its submodels are assigned a unique identifier (IRDI or URI) for global location and referencing. Metadata provides a high-level description to facilitate discovery.
- Security and Access Control: Includes mechanisms to define which stakeholders (for example, technician, supplier, customer) are authorized to view or modify specific parts of the AAS.

TRY IT OUT: BROWSE THE ASSET ADMINISTRATION SHELL

You can try out an online AAS browser that contains entries for a range of equipment at https://v3.admin-shell-io.com/. Figure 5.10 shows an example of a Siemens pressure gauge showing it’s nameplate sub model and identifier.